UTHEALTH HOUSTON AND MEMORIAL HERMANN HIGH RISK LIVER TRANSPLANT PROTOCOL FOR PERIHILAR CHOLANGIOCARCINOMA V7

Preserve the established neoadjuvant and transplant pathway while separately evaluating larger radial tumors, previous primary-tumor biopsy, previously positive nodes and regimen changes. A high-risk feature within standard eligibility is not the same as an eligibility expansion.

Version 7 | Revised 2026-10-24 06:55 CDT | Institutional approval pending

POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

1 PURPOSE AND PATIENT SELECTION

1.1 IDENTIFY THE STANDARD PATHWAY BEFORE CONSIDERING EXPANSION

- Confirm the lesion is perihilar, not a distal bile-duct cancer or an intrahepatic primary. Document a malignant-appearing stricture on cholangiography plus the applicable diagnostic evidence: malignant cytology/biopsy, CA19-9 above 100 U/mL without cholangitis, aneuploidy, or the qualifying hilar mass. A negative brushing does not by itself exclude malignancy. [1]
- If a mass is present, standardized exception policy requires a single mass strictly less than 3 cm in radial diameter, measured perpendicular to the duct. Do not substitute longitudinal ductal extent or a less-than-or-equal-to threshold. A mass exactly 3.0 cm is outside that standardized criterion. [1]
- Obtain independent HPB review of technical resectability, vascular reconstruction, future liver remnant and underlying liver disease. Policy requires unresectability from technical considerations or liver disease; a resectable de novo tumor does not acquire a conventional transplant indication merely because an LT series reports favorable survival. [1] [3]

1.2 DEFINE EACH HIGH RISK FEATURE SEPARATELY

- Record radial size, de novo versus PSC-associated disease, CA19-9 after drainage and infection control, prior treatment, node station and pathology, primary biopsy route, and all evidence of metastasis. A CA19-9 above 500 U/mL is a risk finding to explain and discuss, not a threshold that qualifies the patient for transplantation.
- An indeterminate node requires resolution, not automatic exclusion or clearance. EUS sampling of a suspicious node is distinct from transperitoneal biopsy of the primary tumor. A negative nodal sample does not replace operative staging. [1] [4]

The 12-center US experience included 287 pathway patients and 214 recipients. Five-year intention-to-treat survival was approximately 53%, while post-LT recurrence-free survival was approximately 65%; these are different endpoints and denominators. Larger masses were associated with worse outcomes. The old 3–5 cm entry band is not adopted as a validated extension. [5]

PCCA-B1 — SIZE, PRIOR BIOPSY AND PRIOR NODES: ONE EXCLUSION OR SEPARATE RESEARCH STRATA?

QUESTION AND COMPARISON

For potential pCCA transplant recipients, should a radial mass ≥ 3 cm, prior transperitoneal primary-tumor biopsy, or previously proven but apparently cleared malignant nodes be accepted as one expanded group or separate investigational strata?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Darwish Murad 2012 [5] Retrospective 12-US-center NAT-to-LT cohort; 287 entered, 71 dropped out, 214 transplanted. Primary source | Full-text PDF Locator: Gastroenterology 2012; 143: 88–98.e 3; Results, patient flow and selection-criteria recurrence analyses; local full PDF. Review depth: Local full PDF reviewed	**Reported result:** From treatment-pathway entry, 2-/5-year OS was 68%/53%; post-LT 2-/5-year recurrence-free survival 78%/65%. Among 21 transplanted patients with masses >3 cm, 5-year RFS was 32% versus 69% for smaller tumors. The combined outside-criteria group had recurrence hazard ratio 2.98. **Do not infer:** Larger tumors were not randomized. Composite outside-criteria effect cannot be assigned separately to size, biopsy or metastasis. Small >3-cm recipient subset; selection and center effects.
Darwish Murad 2012 [6] Mayo 1993–2010; 199 protocol entrants, 62 dropouts, 131 local LT and six elsewhere. Primary source | TMC: retrieve PDF Locator: Hepatology 2012; 56: 972–981; Tables 2–3 and Transplantation/Recurrence sections. Review depth: Complete publisher HTML relevant methods/results/Tables 2–3 reviewed; PDF pending.	**Reported result:** Dropout predictors included >3-cm mass (adjusted HR 2.14, 95% CI 1.08–4.24), CA19-9≥ 500 (HR 2.29), positive/suspicious intraluminal sampling (HR 3.63) and MELD ≥ 20 (HR 3.48). Local recipients had 26 recurrences; 5-year OS 71%, RFS 68% from LT. **Do not infer:** Strong overlap with Murad multicenter and Heimbach cohorts; not independent replication. These are prognostic associations, not validated universal exclusion thresholds.
Heimbach 2011 · biopsy [4] 191 NAT-to-LT protocol entrants; 16 underwent primary-tumor TP-FNA. Primary source | Full-text PDF Locator: HPB 2011; 13: 356–360; Results and biopsy/staging table; local complete PDF. Review depth: PMC full text reviewed	**Reported result:** Peritoneal metastases at staging: 5/6 with positive TP-FNA versus 14/175 without TP-FNA; 9 negative TP-FNA patients had none; one result was equivocal. **Do not infer:** Only six positive biopsies. Nonrandom sampling and more obvious tumor may confound association. Negative-biopsy subgroup is far too small to establish safety or equivalence.

Study • population • source	Results • interpretation boundary
Aveson 2022 [7] MSK retrospective planned-resection cohort 1991–2014; 262 patients, 37 TP biopsy versus 225 intraluminal/no biopsy. Primary source | TMC: retrieve PDF Locator: HPB 2022; 24: 942–949; Methods, Results and cumulative-incidence analysis. Review depth: Indexed primary full-text methods/results inspected; primary abstract corroborated. Download attempt returned HTML, not PDF; complete PDF still needed.	**Reported result:** Peritoneal metastases at surgery 5.4% versus 7.6%; among 243 initially PM-negative patients, 2-year cumulative PM 20.3% versus 20.1%(p=.7). **Do not infer:** Direct counterevidence to universal biopsy-seeding claims, but shorter exposure interval, no LT waiting/immunosuppression comparison and only 37 exposed patients. Does not overturn allocation policy.
Soliman 2025 · GemCis [8] Houston recipient-only NAT comparison; 27 LT, 26 analyzed after one treatment-intolerance exclusion. Primary source | Full-text PDF Locator: Transplant Direct 2025; 11: e 1760; Tables 2–4, Recipient Posttransplant Outcomes, Figure 2. Review depth: PMC full-text sections reviewed	**Reported result:** Occult positive nodes in 6/26(23.1%); nodal disease associated with worse OS and RFS. Nine recurrences overall; main-text 1-/3-year OS 83.0%/52.3% from LT. **Do not infer:** Occult nodes discovered after LT are not evidence supporting knowingly positive nodes cleared by therapy. No all referred or all-treated denominator. Abstract/body OS rounding differs; use full Results values.
Dong 2026 · resection versus LT [3] Mayo de-novo pCCA 1993–2023; 127 analyzed LT-pathway entrants, 52 dropouts, 75 LT. Primary source | Full-text PDF Locator: Hepatology 2026; institutional protocol, patient-flow Figure 1 and Results; local full PDF. Review depth: Complete publisher PDF reviewed, including institutional protocols and intention-to-treat comparison	**Reported result:** Dropouts included 15 with regional nodal metastases and 34 with distant metastases; three medical deterioration. **Do not infer:** Provides the consequence of detecting nodes, not outcomes of clearing them and transplanting. Overlaps historical Mayo cohorts.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Size, biopsy exposure and nodal biology must remain separate. Larger-mass LT has directly unfavorable recurrence data. Biopsy evidence is conflicting across settings, so the mechanism is plausible and current policy is restrictive, but an inevitable-seeding claim is unjustified. No reviewed study validates a cleared-node rescue stratum. Imaging-negative nodes do not equal pathological clearance. [5] [6] [4] [7] [8] [3]

- Aveson challenges a universal causal inference from Heimbach. Absence of PM in nine negative TP-FNA cases does not establish safety. Soliman's occult-node recipients do not validate deliberate nodal expansion. The exact 3.0-cm boundary is a current policy boundary, not a biological discontinuity proven by these studies.

Certainty / applicability: Moderate confidence that conventional selection yields better observed outcomes; low certainty for causal effects of size and biopsy; very low/no direct evidence for a previously positive-node clearance pathway.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- POLICY: OPTN 9.5. A specifies radial mass < 3 cm, unresectability, NAT and staging; primary TP-biopsy must be avoided. STUDY: >3 cm associated with lower RFS; positive TP-FNA associated with PM in a very small LT cohort. LOCAL: closing all prior-biopsy and cleared-node strata is a conservative first-study choice, not proof of universal futility.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
OPTN policy [1] "single and less than three cm in radial (perpendicular to the duct) diameter." Policy 9.5.A, printed p191; effective 1 October 2026 Source | Full-text PDF	Binding standardized-exception mass wording; historical ≤3 cm study definitions are not identical.
Heimbach 2011 · biopsy [4] "5/6 (83%) had peritoneal metastases at operative staging." Published PDF p1 / journal p356, abstract Results Source | Full-text PDF	This was the diagnostic-positive primary-FNA subgroup, not all biopsy-exposed patients.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Keep size expansion, primary transperitoneal-biopsy exposure and cleared malignant nodes as three separate closed transplant strata. [1] [4] [5] [6] [7] [8] [3]

- For the conventional standardized-exception mass criterion, retain a single mass <3.0 cm in radial diameter.
  - Exactly 3.0 cm is not rounded down; obtain central measurement review.
  - Do not activate a 3–5 cm expansion.
- Exclude any prior transperitoneal primary-tumor biopsy exposure from the initial transplanting study.
  - Record trajectory and whether pathology was positive, negative or nondiagnostic; this is a conservative selection rule, not proof of dissemination in every exposed patient.
- Exclude previously proven malignant nodes from the initial transplant cohort even after radiographic/pathologic clearance.

- Nodal EUS sampling that does not traverse the primary is distinct from primary-tumor biopsy.
- Do not stack these adverse strata.

Why retain or change it: OPEN Analyst A proposal: initial study retains a single radial mass < 3.0 cm and no nodal/extrahepatic disease. Do not open a 3–5 cm arm. Keep prior primary TP-biopsy exposure outside the initial arm, recorded as positive, negative or nondiagnostic rather than called inevitably disseminated. Keep previously proven malignant nodes outside the initial arm even when imaging clears. Permit EUS sampling of suspicious regional nodes when the needle does not traverse the primary tumor. An unexpected prior biopsy triggers specialist/policy review, not an assertion that every transplant is legally prohibited.

INFORMATION REQUIRED BEFORE THE DECISION

Original MRI/MRCP or CT with perpendicular radial measurement; full biopsy reports/needle trajectory; node station and original pathology; operative staging plan; resection assessment. Record whether EUS sampled a regional node or traversed the primary mass, the route, pathology result and date. These are not equivalent exposures.

Clinical versus allocation: Clinical review, standardized exception eligibility and research expansion are separate. A favorable response or living donor does not erase the original size/biopsy/node history.

Roadmap / protocol reconciliation: Preserve the pCCA roadmap's early transplant consultation and primary-biopsy protection. Reconcile the roadmap's stale summary that calls pCCA nonstandardized: Policy 9.5.A provides a standardized pathway for qualifying cases.

Meeting record: pCCA-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

PCCA-B2 — WHEN IS RESECTION A GENUINE COMPETING CURATIVE OPTION?

QUESTION AND COMPARISON

For de-novo pCCA potentially amenable to R 0 resection, does LT improve net outcomes enough to replace optimized resection, and how should marginal FLR or reconstructible vascular involvement affect the route?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Dong 2026 · resection versus LT [3] Single-center 1993–2023 de-novo pCCA; 288 intended resection (207 resected, 81 dropout) and 127 intended NAT/LT(75 LT, 52 dropout). Primary source | Full-text PDF Locator: Hepatology 2026; 1128–1140; Methods Statistics, Figure 1 and OS/ITT analyses; PDFpp 3–8. Review depth: Complete publisher PDF reviewed, including institutional protocols and intention-to-treat comparison	**Reported result:** As-treated median OS: LT 78 months, resection without vascular resection 58.2, with vascular resection 25.8; 5-year 58.6%, 49.6%, 26.7%. ITT median OS 38.5 months LT versus 31.7 resection (p=.19). **Do not infer:** Not equivalence and not proof resection is superior. Historical era, selection and dropout differ. Methods measure LT time from NAT enrollment but resection time from surgery, so even ITT clocks are not perfectly aligned; as-treated LT percentages must not be called purely post-LT.

Study • population • source	Results • interpretation boundary
Ethun 2018 [9] Multicenter 2000–2015; 234 attempted resections and 70 entered LT protocols; 191 curative-intent resections and 41 confirmed-CCA LT in recipient analysis. Primary source | TMC: retrieve PDF Locator: Ann Surg 2018; 267: 797–805; primary abstract and indexed Results/Discussion. Review depth: Primary abstract plus indexed full-text excerpts; PDF, exact ITT analytic exclusions and time-origin detail require final full-PDF check.	**Reported result:** Recipient 3-/5-year OS 72%/64% LT versus 33%/18% resection. Restricted non-PSC,<3 cm, node-negative comparison favored LT at 5 years 54% vs 29%(p=.03); adjusted/restricted ITT comparison remained significant (p=.049). **Do not infer:** Conflicting observational evidence versus Dong; major selection/era/therapy confounding. Exact ITT exclusions and time origin require full-PDF recheck; no unsupported ITT rate is reproduced here.
Vugts 2021 [10] 732 consecutive Dutch pCCA presentations 2002–2014; retrospective application of Mayo eligibility. Primary source | Full-text PDF Locator: Ann Surg Oncol 2021; 28: 1483–1492; Methods Eligibility/Statistical Analyses and Results; complete 10-pagePDF. Review depth: Full publisher article and complete 10-page PDF verified; correction link recorded.	**Reported result:** Without LT, median survival 13 months (95% CI 3–23) from diagnosis/presentation. This shows unmet need among unresectable selected patients, not benefit over R 0 resection. **Do not infer:** Historical non-LT care; no matched LT comparator, suspected/confirmed nodes pooled and infection criterion incompletely reconstructible. Abstract 24 PSC versus body 29 PSC inconsistency is irrelevant to the decision and not propagated.
Olthof 2020 [11] 20-center retrospective 1667 patients; 1484 combined liver/bile-duct resections analyzed; 298 PVE. Complete-data propensity analysis 98 PVE versus 98 controls. Primary source | Full-text PDF Locator: Ann Surg Oncol 2020; 27: 2311–2318; Methods, Matched Cohort and Table 3; complete 8-pagePDF. Review depth: Full publisher article and complete 8-page PDF verified; matched analysis and exclusions checked.	**Reported result:** Matched PVE versus noPVE: ISGLS B/C liver failure 8% vs 36%; 90-day mortality 7% vs 18%; bile leak 10% vs 35%. Median remnant-volume gain 42% after 22 days. **Do not infer:** Unmatched outcomes were worse in PVE patients because they were higher risk. Matching omits unmeasured confounding and excludes missing volumetry; does not prove one FLR cutoff. Operated-only denominator omits patients failing to reach surgery.

Study • population • source	Results • interpretation boundary
Darwish Murad 2012 [5] 287 NAT/LT entrants across 12 US centers; 214 transplanted. Primary source | Full-text PDF Locator: Gastroenterology 2012; Results and survival figures; full PDF. Review depth: Local full PDF reviewed	**Reported result:** 5-year pathway-entry OS 53% and post-LT RFS 65% are useful feasibility benchmarks, but answer a different question from randomized LT-versus-resection benefit. **Do not infer:** Substantial overlap with Mayo comparisons; do not count as an independent comparative test.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Evidence supports LT as a serious option for carefully selected unresectable pCCA. It does not resolve LT versus feasible optimized R 0 resection. Ethun favors LT even after restricted ITT analysis, whereas Dong loses statistical advantage when dropout is counted. Neither randomization nor comparable entry clocks remove the confounding. PVE can materially change the surgical risk assessment. [3] [9] [10] [11] [5]

- Do not cite Dong's nonsignificant ITT p-value as equivalence or proof that resection must always be better. Conversely, recipient-only LT survival cannot erase donor opportunity cost, NAT dropout and wait-list hazards. Reconstructible vascular involvement or a correctable small FLR alone is not proof of permanent unresectability.

Certainty / applicability: Low comparative certainty; stronger but still nonrandomized evidence for FLR optimization before declaring resection unsafe.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- POLICY: standardized pCCA exception requires unresectability or underlying liver disease. STUDY: PVE growth was assessed at median 22 days in the matched study; comparisons remain observational. GUIDANCE: CIRSE supports higher remnant targets in cholestatic/cirrhotic liver. LOCAL: >40% planning target, reassessment 2–4 weeks and required expert review are workflow proposals; inadequate function/infection can prevent surgery despite adequate volume.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Dong 2026 · resection versus LT [3] "If the future liver remnant was insufficient, preoperative portal vein embolization was conducted to enhance hepatic reserve." Published PDF p3 / journal p1130, institutional protocols Source | Full-text PDF	Resection optimization is part of the comparator, not a failure of transplant selection.
Bilhim 2024 [12] ">40% for cirrhotic or cholestatic livers" CIRSE 2024, Table 2; spacing normalized Source | Full-text PDF	Specialist practice guidance supporting a planning target, not a guaranteed safety threshold.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use resection first when an R0 operation and an adequate functional remnant are feasible. [3] [12] [9] [10] [11] [5]

- Proposed planning target for cholestatic/cirrhotic liver: FLR/TLV >40%, combined with function, drainage and infection assessment.
  - Volume alone never establishes operative safety; 40% is not an OPTN requirement.
- If insufficient remnant is the only barrier, assess selective drainage and augmentation/PVE; reassess remnant volume and function about 2–4 weeks after PVE.
  - Vascular reconstruction or PVE need alone does not prove unresectability.
- Return to LT eligibility review only if an R0 resection remains infeasible or unsafe after reasonable optimization.
  - Do not transplant otherwise resectable de novo pCCA outside a specifically approved comparative study.

Why retain or change it: OPEN: resection remains the default when an expert HPB review judges R 0 achievable with adequate functional remnant. Obtain contrast liver imaging/MRCP, vascular reconstruction plan, volumetry, bilirubin/cholangitis status and functional assessment when volumetry is marginal. For cholestatic/cirrhotic livers aim for FLR>40% as a planning target, not automatic clearance. Drain the intended remnant as clinically indicated, control infection and consider PVE; repeat volumetric/functional assessment around 2–4 weeks and document growth. LT referral is appropriate for persistent technical/functional unresectability. Resectable de-novo pCCA belongs in a prospective comparative protocol, not routine substitution.

INFORMATION REQUIRED BEFORE THE DECISION

Two specialist HPB/transplant reviews; vascular/ductal map; CT volumetry; functional assessment where available; cholangitis/drainage status; PVE response and reconstruction plan.

Clinical versus allocation: Policy 9.5.A requires unresectability from technical considerations or underlying liver disease; comparative recipient survival alone does not satisfy this requirement.

Roadmap / protocol reconciliation: The roadmap leaves numerical remnant criteria for specialty approval. Record this as a proposed local addition, with individualized function-based exceptions documented.

Meeting record: pCCA-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

2 PRETREATMENT WORKUP AND BIOPSY PROTECTION

- Obtain or review contrast liver MRI/MRCP and/or multiphasic CT for ductal and vascular extent, chest imaging, CA19-9 with bilirubin/infection context, CBC, renal and liver chemistry, coagulation, functional status, nutrition and transplant medical/psychosocial assessment. [1] [13]
- Plan biliary drainage with advanced GI and the HPB/transplant team. Treat cholangitis before interpreting CA19-9 or initiating an intensive regimen. Record stent/drain location, cultures when indicated and subsequent drainage strategy.

- Use ERCP-based brush cytology, FISH/aneuploidy testing or appropriately planned intraductal sampling when needed. Avoid transperitoneal aspiration or biopsy of the primary by EUS, percutaneous or operative routes when LT is contemplated. If already performed, preserve the original report, trajectory and pathology and hold the standard route for specialist review. [1] [4]
- Stage suspicious regional or distant nodes by a route that does not traverse the primary tumor. Consider PET/CT for additional metastatic assessment when it can change management, but do not regard a negative PET as proof that nodes or peritoneum are sterile.
- For the standardized exception, document chest and abdominal cross-sectional staging within 90 days before the initial request; extension imaging must be within 30 days before submission. Operative nodal/peritoneal staging is required after neoadjuvant therapy and before LT. [1]

Advance to treatment planning only after diagnosis, resectability and original staging are established and the exact intended pathway is identified. Hold for unresolved nodes, incomplete biopsy history, uncontrolled infection or unapproved expansion. Confirmed nodal, intrahepatic metastatic or extrahepatic metastatic disease prevents progression through the conventional protocol. [1]

3 NEOADJUVANT THERAPY AND BILIARY CARE

3.1 USE ONE DEFINED TREATMENT PROGRAM

- Use the center's approved written pCCA neoadjuvant program with oncology, radiation oncology and transplant-surgical sign-off. Specify actual drugs, dose modifications, radiation fields/fractionation, drainage management and staging sequence. Policy requires neoadjuvant therapy and operative staging but does not mandate one universal drug or radiation recipe. [1] [5]
- Do not automatically replace the established neoadjuvant program. The OPEN pCCA-B3 review now considers both SBRT/capecitabine and a separately governed GemCis-only feasibility/comparative arm; neither is established as equivalent to the Mayo-style pathway. ICI-containing induction or simultaneous boundary expansions are not activated.

The previous statement that only the Mayo hyperfractionated regimen has LT outcome data was too strong. Small SBRT/capecitabine and multiagent-systemic cohorts exist, but they do not establish noninferiority or make regimens interchangeable. Any alternative needs a defined prospective treatment and toxicity plan. [14] [8]

In a 49-recipient observational comparison, neoadjuvant chemoradiotherapy was associated with reduced recurrence but no demonstrated OS advantage. Major vascular complications occurred in 11/27 versus 3/22; hepatic-artery complications were 9/27 versus 2/22. Center, selection and regimen differences limit causal interpretation. The recurrence signal must be weighed against graft and vascular harm. [15]

PCCA-B3 — MAY A NON-MAYO NEOADJUVANT REGIMEN REPLACE THE CONVENTIONAL PROGRAM?

QUESTION AND COMPARISON

For otherwise conventionally selected pCCA LT candidates, which NAT regimen should be the default and which alternatives deserve prospective evaluation, given control, dropout and vascular/biliary harm?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Darwish Murad 2012 [5] 287 entrants/214 LT across 12 US centers. Primary source | Full-text PDF Locator: Gastroenterology 2012; Treatment Characteristics and Outcomes; full PDF. Review depth: Local full PDF reviewed	**Reported result:** 5-year entry OS 53%; post-LT RFS 65%. No clear RFS difference by brachytherapy receipt was detected. **Do not infer:** Strongest mature multi-center pathway benchmark, but no randomized regimen comparison; lack of detected brachytherapy effect does not establish interchangeability.
Welling 2014 · SBRT [14] Single-center pilot; 17 evaluated, 12 NAT, 9 listed, 6 LT. Primary source | Full-text PDF Locator: Liver Transpl 2014; 20: 81–88; Methods, patient flow and adverse-event tables; local PDF. Review depth: Indexed full-text content reviewed	**Reported result:** SBRT 50–60 Gy in 3–5 fractions over 2 weeks, followed by capecitabine; 6 LT with 1-year OS 83%.35 pre-LT adverse events and 11 post-LT complications were reported. **Do not infer:** Very small feasibility cohort, short oncologic follow-up, no randomized comparator. Regimen is a study description, not a universal radiation prescription or proof SBRT is the preferred alternative.
Soliman 2025 · GemCis [8] Recipient-only retrospective 2008–2023; 26 analyzed LT, 12 GemCis-only, 14 RT-based including six also receiving GemCis. Primary source | Full-text PDF Locator: Transplant Direct 2025; 11: e 1760; Tables 1–4, Figure 2, Discussion; full PDF. Review depth: PMC full-text sections reviewed	**Reported result:** Main-text 1-/3-year OS 83.0%/52.3%; 9/26 recurrences; 1-/3-year RFS 77.8%/59.0%. No statistically significant OS or explant-response difference between treatment strata; RT had more operative blood-loss/transfusion signals. **Do not infer:** No pretreatment denominator or differential dropout data; small nonrandom groups, era/selection confounding and occult nodal disease. Supports studying omission, not equivalence. Use main-text values rather than abstract 84%/55%.
Hoogwater 2023 · radiation [15] Multicenter 2011–2020; 49 LT, 27 with NAT, 22 without. Primary source | Full-text PDF Locator: BJSOpen 2023; 7: zrad 025; Table 3, adjusted recurrence analysis and mortality discussion; full PDF. Review depth: Full text reviewed	**Reported result:** Post-LT 5-year OS 41% NAT versus 53% no NAT(not significant); NAT associated with less recurrence (adjusted HR 0.30, 95% CI 0.09–0.97). Major vascular complications 11/27 vs 3/22; hepatic-artery complications 9/27 vs 2/22. **Do not infer:** Competing recurrence benefit and treatment harm, not proof RT can routinely be omitted. Abstract 9 vs 2 vascular wording should not replace Table 3's broader 11 vs 3 major-vascular count.

Study • population • source	Results • interpretation boundary
Ito 2025 · LDLT [16] Prospective Kyoto pilot; 10 enrolled, 5 LDLT, 5 dropout. Primary source | TMC: retrieve PDF Locator: J Hepatobiliary Pancreat Sci 2025; 32: 276–286; patient flow, Tables and Discussion; full HTML. Review depth: Publisher full HTML methods/results inspected; complete PDF not found locally.	**Reported result:** All 5 recipients alive at 1 year, median follow-up 23.7 months; one recurrence and 2/5 hepatic-artery thromboses. Dropouts included progression, positive nodes, toxicity/complication, financial withdrawal and one conversion to resection. **Do not infer:** Five recipients cannot establish long-term oncologic safety; different allocation/LDLT setting and demanding regimen. This is not support for adding immunotherapy.
Darwish Murad 2012 [6] 131 local LT within 199 protocol entrants. Primary source | TMC: retrieve PDF Locator: Hepatology 2012; Transplantation and Recurrence sections. Review depth: Complete publisher HTML relevant methods/results/Tables 2–3 reviewed; PDF pending.	**Reported result:** 11/131 required retransplantation, including six for HAT; 5-year post-LT OS 71%, RFS 68%. **Do not infer:** Illustrates clinically significant graft risk; not an independent estimate additive to the multicenter series.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Conventional fluoropyrimidine-based chemoradiation remains the most mature benchmark. Alternatives have genuine LT evidence, but none has demonstrated noninferiority. The evidence does not justify singling out SBRT as the only acceptable research alternative while categorically closing GemCis-only. Vascular/biliary harm needs prospective surveillance alongside recurrence and dropout. [5] [14] [8] [15] [16] [6]

- RT may reduce recurrence yet increase vascular morbidity. Chemo-only favorable selected-recipient outcomes do not reveal who progressed before LT. No-NAT European recipients are not a rationale to omit all induction in a US exception pathway. GCS/RT and SBRT are not interchangeable.

Certainty / applicability: Moderate confidence in feasibility of conventional selected NAT/LT; very low comparative certainty between named regimens. Direct vascular safety signal deserves action despite nonrandomization.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- POLICY: OPTN requires an approved written protocol including NAT/staging but does not mandate a particular radiation dose. STUDY: SBRT 50–60 Gy/3–5 fractions and GCS/56 Gy are published pilot recipes only. LOCAL: GemCis-only prospective arm, no simultaneous biology expansion, and initial ICI exclusion are risk-management proposals. No validated minimum cycle count or safe vascular dose threshold is inferred.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Welling 2014 · SBRT [14] “treated with SBRT (50–60 Gy, 3–5 fractions over two weeks)” Published PDF p1 abstract; protocol details p3 Source | Full-text PDF	Small pilot experience supports study development, not noninferiority or interchangeability.
Hoogwater 2023 · radiation [15] “associated with a higher rate of early hepatic vascular complications” Published PDF p1, abstract Conclusion Source | Full-text PDF	The recurrence signal must be weighed against vascular/graft harm; observational comparison is confounded.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Keep the established Mayo-style pathway as default; develop GemCis-only and SBRT/capecitabine as separately governed alternatives, without claiming equivalence or superiority. [14] [15] [5] [8] [16] [6]

- Default to the established fluoropyrimidine/EBRT-based pathway under the approved radiation and pharmacy protocol.
  - Do not treat a pilot dose/fractionation or cycle count as a universal order.
- Develop a separately governed prospective GemCis-only feasibility/comparative arm for otherwise conventionally selected pCCA.
  - Retain SBRT/capecitabine as another documented pilot option, not the uniquely evidence-preferred alternative.
  - Neither arm is activated by this document.
- Do not combine either alternative with an initial size, primary transperitoneal-biopsy or previously malignant-node expansion.
  - Keep ICI-containing induction outside these initial arms unless separately approved on pre-LT safety evidence.
- Before opening, specify regimen delivery, staging, stopping rules, radiation-field and arterial/biliary reconstruction review.
  - Capture every entrant, failure to reach LT, cholangitis, HAT/PVT, biliary injury, graft loss, recurrence and all graft use.

Why retain or change it: OPEN revised stance: use the existing named Mayo-style fluoropyrimidine/EBRT pathway as default, with boost technique and capecitabine specified by the approved radiation/pharmacy protocol. Develop GemCis-only as a separately governed prospective alternative arm for otherwise conventional biology; do not substitute it routinely or label it equivalent. Retain SBRT/capecitabine as another

documented pilot option, but not the uniquely preferred alternative. No simultaneous size, primary-biopsy or cleared-node expansion in either arm. Keep ICI-containing induction outside these initial arms pending a separate pre-LT ICI safety protocol. Record all entrants, intended regimen, failures to reach LT, cholangitis, graft loss, HAT/PVT and recurrence.

INFORMATION REQUIRED BEFORE THE DECISION

Written radiation/chemotherapy plan and dose modifications; drainage/infection plan; biliary/vascular anatomy; actual prior dose maps; operative staging and graft-safety endpoints.

Clinical versus allocation: Policy requires a written neoadjuvant protocol and operative staging, not this exact regimen. A regimen study does not relax the standardized tumor criteria.

Roadmap / protocol reconciliation: Use the biliary roadmap for supportive/systemic care. Do not import advanced-disease GemCis–ICI efficacy as proof of pre-LT safety.

Meeting record: pCCA-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

3.2 ICI AND OTHER INVESTIGATIONAL COMPONENTS

- If prior ICI exposure occurred, record drug, last dose, immune-related toxicity and its treatment. Arrange transplant-hepatology, oncology and pharmacy review. An approximately 90-day interval may be proposed as a planning safeguard, but neither 50 nor 90 days is a universally safe or CCA-validated cutoff. Record residual rejection/graft-loss uncertainty. [17] [18] [19]
- The small mixed HCC/CCA recipient series does not validate routine ICI bridging. Although its CCA results reported no rejection, recurrence or death, a CCA recipient had immediate graft failure requiring retransplantation. Selected recipients and inconsistent case descriptions cannot establish safety of a four-week washout. [19]
- Do not mandate serial ctDNA, Y90 or photodynamic treatment to qualify for LT. A new liver-directed component requires an explicit rationale, arterial/biliary feasibility assessment and prospective safety oversight. ctDNA is an exploratory adjunct, not an independent proceed or stop rule.
- Continue maintenance therapy only under the selected oncology program, with dose and stop date reconciled with surgery, marrow recovery, infection, renal function and any ICI interval. Do not use a generic “until transplant” instruction without a perioperative plan.

4 RESPONSE ASSESSMENT AND RESTAGING

- Proposed local schedule for approval: review symptoms, drainage/infection and laboratory toxicity during each treatment cycle; repeat cross-sectional disease assessment at phase transitions and approximately every 8–12 weeks during prolonged therapy. Update chest/abdominal staging and CA19-9 before final selection and according to exception submission windows. [1]
- Interpret CA19-9 after relief of obstruction and infection, with nonsecretor status considered. Rising CA19-9 or new symptoms trigger review and confirmatory assessment; they do not substitute for anatomical staging.
- Advance to operative staging only with controlled disease, no new metastatic site, acceptable medical/functional reserve, adequate biliary drainage, a completed treatment and toxicity record, and a documented eligible clinical or approved study route.

- Hold for equivocal nodes, possible peritoneal disease, discordant imaging/markers, unresolved toxicity or unapproved regimen deviation. Progression, proven metastatic nodes, intrahepatic metastases or extrahepatic disease stops the conventional route. [1]

5 OPERATIVE STAGING AND TRANSPLANT CLEARANCE

- After neoadjuvant treatment and before LT, inspect the peritoneum and assess regional hepatic nodes. Sample suspicious sites and obtain pathology. Specify staging laparoscopy or laparotomy in the surgical plan. Additional nodal stations or washings may be used according to findings; an indiscriminate extended dissection is not an automatic requirement. [1] [5]

- Advance to LT only when staging excludes regional nodal, intrahepatic metastatic and extrahepatic disease, the patient remains fit, technical reconstruction is feasible and all clinical, allocation and donor approvals are complete. Positive malignant staging stops this pathway; indeterminate pathology keeps it on hold. [1]

- The operative plan must address irradiated hepatic artery/portal structures and biliary reconstruction. Retain actual prior radiation and arterial-therapy records. Record arterial/biliary complications and any additional graft use. [15]

- Before recipient activation, document the intended DDLT or LDLT route, actual exception decision if applicable, expected waiting time, and a recipient-benefit assessment against contemporary non-LT care. Do not promise priority or treat DCD, an extended-criteria graft or machine perfusion as evidence that no other candidate could benefit. [1] [13]

- For LDLT, retain independent donor advocacy and donor-team assessment, freedom to withdraw, and separate consent covering uncertain recipient benefit, operative and long-term morbidity, recovery, financial burden and possible retransplantation. Do not relax donor safety criteria or make possession of a living donor a clinical eligibility requirement. [1] [13] [20]

- The living-donor literature reports low but nonzero mortality and incomplete recovery follow-up. Use current center- and graft-specific counseling; do not pool overlapping registry studies or portray donor survival as proof of complete recovery. [20]

ALLOC-B1 — WHAT ARE WE APPROVING?

Evidence and uncertainty before the shared rule

The separation is a governance requirement, not an empirical survival cutoff. A clinically appropriate patient can lack exception eligibility, and an approved exception does not prove clinical benefit. LDLT requires an independent donor determination; a registry entry is not permission to perform experimental transplantation. [1] [2] [21] [22]

Evidence-to-threshold distinction: Six domains: local checklist architecture, not six mandatory national approvals.; 100% applicable completion: governance safeguard, not an outcome-derived percentage.; Hold pending required authorization: operational implementation of separate clinical, donor and research responsibilities.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require 100% completion of the applicable clinical, allocation, donor and research determinations before activation. [2] [1] [21] [22]

- Complete six separate dated determinations: clinical suitability; standardized-exception eligibility; nonstandard NLRB review status; calculated-MELD access; LDLT donor clearance; and research authorization.
  - Mark nonapplicable domains explicitly with a reason—do not manufacture six approvals for every patient.
- Any required missing, pending or adverse determination is a HOLD.
  - An exception application is not approval, a donor is not recipient benefit, and clinical approval is not study activation.
- Recheck the current policy/guidance version before each request; record actual decision and priority rather than predicted entitlement.

Why retain or change it: Retain 100% completion of applicable determinations with dated responsible owners and reasoned N/A entries. A required unresolved determination means HOLD, not an assumed refusal or an authorization by default.

INFORMATION REQUIRED BEFORE THE DECISION

Signed suitability record; exact allocation route and policy version; exception outcome; donor-team status; study and institutional activation record.

Clinical versus allocation: Policy 9.4 permits discretionary requests where appropriate; absence of favorable diagnosis-specific guidance is not absence of a review mechanism, and a request does not promise approval.

Roadmap / protocol reconciliation: Use the same clinical-versus-allocation separation in every standard and high-risk pathway.

Meeting record: ALLOC-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B2 — WHAT IS THE RELEVANT ORGAN OPPORTUNITY COST?

Evidence and uncertainty before the shared rule

Use pathway-level benefit from an agreed entry date, not recipient-only survival. Compare with contemporary tumor-specific care and actual local access. LDLT avoids one direct deceased-donor allocation but creates healthy-donor risk and may generate rescue graft use. Graft quality/perfusion changes do not make opportunity cost zero. [22] [1] [23] [20] [24] [25] [26]

Evidence-to-threshold distinction: Two reviewers: local independence safeguard, not validated optimum.; 6/12/18 months: sensitivity scenarios for access/dropout, not mandatory waits.; HCC ≥60% five-year ITT OS: one expert trial-design benchmark; not a cross-cancer allocation policy.; All grafts: count initial and rescue/retransplant grafts; donor morbidity remains a separate outcome.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Require two independent recipient-benefit reviews and an explicit organ-use case; do not adopt a universal five-year survival cutoff. [25] [20] [22] [1] [23] [24] [26]

- Proposed gate: two reviewers not both responsible for recruitment must document expected incremental benefit over the actual contemporary alternative, with unresolved disagreement referred to the full committee.
- For DDLT, show local expected access/dropout and sensitivity scenarios at 6, 12 and 18 months, total graft use including retransplantation, and plausible alternative recipients.
  - These are planning scenarios, not imposed waiting periods.
- For LDLT, separately show healthy-donor risk and potential downstream deceased-donor graft use.
  - Do not call DCD, extended-criteria grafts or machine perfusion "zero opportunity cost." No priority changes follow from an AI-selected survival threshold.

Why retain or change it: Retain independent benefit and organ-use reviews, but require disease-specific ITT evidence and explicit uncertainty. No universal five-year survival cutoff or numerical allocation weight is adopted.

INFORMATION REQUIRED BEFORE THE DECISION

Same-eligibility/time-zero comparator; expected access; dropout; donor/graft type; uncertainty intervals; graft loss/retransplant estimate; written independent reviews.

Clinical versus allocation: Only the applicable allocation process determines priority. Favorable post-LT prognosis, evidence grade and urgency are not interchangeable measures.

Roadmap / protocol reconciliation: Compare against the cancer roadmap's actual resection, ablation, HAI, radiation or systemic alternative—not historical untreated survival.

Meeting record: ALLOC-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

ALLOC-B3 — WHAT MUST DONOR CONSENT AND FOLLOW-UP INCLUDE?

Evidence and uncertainty before the shared rule

Donor mortality is low but not zero; complication, recovery and financial burdens are more frequent. Registry missingness and selected healthy donors limit reassurance. Consent must include uncertain recipient benefit, recurrence and possible early graft loss, without pressure to proceed or relax donor criteria. [20] [27] [28] [29] [1]

Quantitative donor context: Kelly reports 8/8,619 deaths by 90 days across eras; A2ALL recorded ≥1 complication in 296/740 donors. These are different outcomes/eras, not poolable rates. The complete donor evidence and follow-up limitations are in ALLOC-B3.

Evidence-to-threshold distinction: Baseline/6 weeks/6 months/1/2 years then annually to five: proposed support schedule; current mandatory reporting remains separately checked.; One sentinel event: conservative review/pause trigger, not a statistically estimated harm boundary.; No relaxed donor criteria: protection against shifting uncertain recipient benefit onto a healthy person.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Do not relax donor criteria; fund independent follow-up and pause after a serious sentinel donor event. [1] [20] [27] [28] [29]

- Proposed readiness gate: independent donor evaluation/advocacy, documented freedom to withdraw, recipient-benefit uncertainty consent, and funded medical, psychological, recovery and financial follow-up before investigational LDLT.
- Proposed local follow-up: baseline, six weeks, six months, one year, two years, then annually through five years.
  - Meet current OPTN reporting duties separately; this five-year support plan is not claimed to be a national reporting requirement.
- One donor death, life-threatening complication, permanent major disability or liver failure requiring transplantation triggers an immediate pause of the relevant investigational LDLT activity and independent review.
  - Other serious donor harm also warrants review; reopening needs independent authorization.

Why retain or change it: Retain independent advocacy, ability to withdraw, funded medical/psychological/financial follow-up and sentinel-event pause. Add explicit denominator/era caveats to consent estimates and collect local outcomes prospectively.

INFORMATION REQUIRED BEFORE THE DECISION

Graft/resection-specific and center-specific risk; donor recovery/employment baseline; independent consent and advocate records; funded follow-up/support; sentinel-event response owner.

Clinical versus allocation: LDLT clearance is separate from recipient selection; the potential need for retransplantation must be included in organ-use discussions.

Roadmap / protocol reconciliation: The same independent donor safeguards apply whether recipient disease is standard or experimental.

Meeting record: ALLOC-B3 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

6 POST TRANSPLANT MANAGEMENT

- Use institutional immunosuppression and infection prophylaxis with transplant-pharmacy review. Prior ICI is not an indication for unvalidated high tacrolimus targets, automatic ATG or scheduled surveillance biopsies. Investigate graft dysfunction promptly and use for-cause biopsy as appropriate. [13] [18]
- There is no established routine post-LT adjuvant cytotoxic, targeted or checkpoint regimen for this high-risk population. Residual tumor or an actionable mutation on explant does not itself authorize preventive systemic treatment. Document an individual plan or a separately approved study.
- Proposed local oncologic follow-up: chest and contrast abdominal imaging with CA19-9 every three months for two years, then every six months through year five, adjusted for risk, renal function and symptoms. Functional imaging and ctDNA are problem-solving or research tools, not fixed mandatory surveillance tests.
- Confirm and restage suspected recurrence. Select local or systemic treatment through a CCA/transplant MDT, considering disease distribution, graft reserve, interactions and patient priorities. Post-LT ICI carries a distinct graft-loss risk and is not routine recurrence therapy. [30]

7 OUTCOMES AND SAFETY OVERSIGHT

- Register every referred or screened candidate in the institutionally approved system, not only recipients. Record acceptance or rejection, treatment entry, response, listing, exception decision, conversion to resection or ablation, progression, toxicity, decompensation, death, patient withdrawal and access-related loss.
- Report overall survival from the prespecified entry point and separately from listing and transplantation. Keep recurrence-free survival, recurrence incidence, graft survival and survival after recurrence distinct, with definitions, confidence intervals, numbers at risk and follow-up completeness. Do not call recipient-only survival an intention-to-treat outcome.
- Capture 90-day recipient morbidity and mortality, biopsy-defined rejection, infection, arterial and biliary complications, graft loss, retransplantation and total grafts used. Record recurrence site, salvage treatment, time off systemic therapy, functional recovery and quality of life, not survival alone.
- Before a transplanting study opens, biostatistics and the independent safety board must approve indication-specific monitoring, evaluable denominators, pause authority and reopening criteria. ALLOC-B4 now proposes explicit initial-cohort review triggers for ratification; they are not validated efficacy/futility boundaries. No universal cross-cancer five-year survival cutoff is adopted.
- Use a contemporary comparator with the same eligibility and time zero. Account for the interval needed to survive to transplantation, crossover, competing risks and overlapping study cohorts. A nonsignificant difference is not proof of equivalence.

ALLOC-B4 — WHAT DATA ARE REQUIRED BEFORE EXPANSION?

Evidence and uncertainty before the shared rule

A small pilot can test feasibility, staging fidelity, data capture and acute safety, not prove durable survival benefit. Define time zero, all-screened enrollment, waitlist failures, post-LT survival/recurrence, competing death, all grafts, donor harms and patient-reported outcomes before opening. An independent board needs actual pause/reopening authority. [23] [31] [21] [20] [32] [33]

Zero observed events among ten fully followed independent patients still has a one-sided exact 95% binomial upper bound of 25.9% (1–0.05^(1/10)). Thus ten survivors cannot establish a <5% event risk. Ninety percent data completeness permits one missing patient; safety events and vital status require active ascertainment, not tolerance of unaccounted loss.

Evidence-to-threshold distinction: Ten per stratum and interim after five at 90 days: local feasibility cap and safety review cadence.; Twelve-month minimum: earliest expansion discussion, not proof of long-term cancer benefit.; ≥90%: proposed non-vital key-measure completeness threshold; not permission to ignore deaths/graft events or unknown status.; One early death/graft loss/re-LT: review/pause trigger, not automatic futility or withholding rescue.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use a small staged pilot with all-screened accounting and explicit safety pauses; no automatic expansion on favorable recipient survival. [23] [31] [21] [20] [32] [33]

- Proposed initial cap: ten transplants per separately approved biological stratum.
  - Review after the first five have 90-day follow-up; no expansion decision until all first ten have at least 12-month follow-up.

  - Reconcile vital status, graft events and known serious harms for every participant; require ≥90% completeness for other prespecified key outcomes.

- In an initial experimental cohort, one all-cause recipient death, graft loss or retransplantation within 90 days triggers a pause of further experimental enrollment/transplantation pending independent review.
  - A later possibly treatment-related death/graft loss or a donor sentinel event also triggers review/pause.
  - Rescue care is never withheld; reopening requires independent authorization.
- These are proposed conservative review triggers, not statistically validated efficacy/futility boundaries.
  - Before opening, biostatistics and the independent board must ratify or replace them with indication-specific rules, denominators, pause authority and reopening criteria.
- Capture every screened candidate, including nonrecipients, conversion to resection, dropout, deaths and access barriers; report survival from entry and from LT, recurrence, graft use, donor recovery and quality of life.
  - Ten survivors alone do not establish benefit or authorize expansion.

Why retain or change it: Retain the small staged pilot as an OPEN governance proposal, not an efficacy claim. Clarify that known safety/vital-status data must be reconciled for every enrolled patient; ≥90% applies only to other prespecified key measures. Indication-specific statistical design may replace the proposed numbers before opening.

INFORMATION REQUIRED BEFORE THE DECISION

Locked eligibility strata; prospective entry/time-zero dataset; modern comparator; adverse-event definitions; follow-up-completeness report; independently signed statistical and safety plan.

Clinical versus allocation: Study monitoring never changes exception criteria or guarantees future access. Audit access by referral, geography, resources and donor availability without turning protected characteristics into exclusion rules.

Roadmap / protocol reconciliation: Retain cancer-roadmap treatment alternatives and document why non-LT care was chosen when candidates do not proceed.

Meeting record: ALLOC-B4 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

8 DECISIONS REQUIRED BEFORE EXPANSION

Resolve pCCA-B1–B3 and ALLOC-B1–B4 in the linked disease-specific meeting record. The immediate protocol correction is to stop treating size, biopsy, cleared nodes and regimen changes as interchangeable. Study activation remains a separate decision.

- Record the selected radial measurement method, handling of exactly 3.0 cm, original biopsy/nodal history, resection comparator, permitted regimen and oversight owners. Transplant surgery, hepatology, advanced GI, HPB surgery, oncology, radiation oncology, pathology and allocation should sign their respective determinations.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

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REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

Problem: The earlier B proposals had an uneven evidence base and emphasized a stance before its supporting and contrary data. **Correction and reason:** Added question-level primary-study tables with selection, denominators, outcome clocks, harms, limitations, exact source excerpts and threshold provenance before each OPEN proposal. Sources/PDF links are adjacent; missing full texts remain flagged.

Problem: Local operational numbers could be mistaken for validated eligibility or allocation thresholds. **Correction and reason:** Separated outcome-derived estimates, guideline/policy wording and local safeguards; revised proposals when the deeper review changed their evidentiary basis. Institutional decisions remain blank.

Approval record Clinical lead _____ Allocation review _____ Research authorization if required _____ Donor review if applicable _____ Date _____